Machine Learning — Lecture 4

Logistic Regression and Classification

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Lecture 4 at a Glance

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Logistic Regression and Classification

Course Learning Outcome

CLO3: Construct and analyze machine learning models by applying algorithmic principles and internal computations.

CLO4: Apply supervised learning techniques to solve practical problems and interpret their outcomes.

Lecture purpose

- Move from regression to classification.
- Learn how logistic regression converts a linear score into a probability.
- Understand binary cross-entropy as the classification loss.
- Compute a forward pass, gradients, and one parameter update.
- Interpret thresholds, coefficients, probabilities, and decision boundaries.

Conceptual structure

Phase	Student question
1. Understand	How is a binary probability model built?
2. Train	How do log loss and gradients improve the model?
3. Use & interpret	How should probabilities guide decisions?
4. Extend & apply	How does the model handle new features and tasks?

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💡 Main idea: logistic regression is a classification model that combines a **linear score**, a **sigmoid probability**, and a **loss function suited to binary labels**.

Learning Objectives

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By the end of Lecture 4, you will be able to:

1. Formulate **binary classification** as a supervised-learning problem.
2. Explain why ordinary linear regression is not ideal for classification.
3. Compute a logistic-regression score $z = w^\top x + b$.
4. Convert a score into a probability using the sigmoid function.
5. Apply a decision threshold to obtain a class prediction.
6. Interpret a linear decision boundary and explain how it shifts when the threshold changes.
7. Compute binary cross-entropy loss for one example.
8. Perform one manual gradient-descent update.
9. Explain threshold trade-offs and probability interpretation.
10. Recognize extensions to multiclass classification.

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Lecture Roadmap

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From class probabilities to reliable decisions

Learning phase	Parts	Central question
1. Understand the logistic model	**I–III**	What is binary classification, how do scores become probabilities, and where is the decision boundary?
2. Train the model	**IV–V**	How do binary cross-entropy and gradients determine the parameters?
3. Use and interpret predictions	**VI–VIII**	How do thresholds, coefficient interpretation, and preprocessing affect decisions?
4. Extend and apply	**IX–X**	How do we handle multiclass outcomes and reason through a real classification case?

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💡 **Practice & consolidation:** compute probabilities and losses, compare thresholds, and identify common modeling mistakes.

Part I

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Classification as Supervised Learning

Classification predicts a **category** or **class label**, rather than a continuous numerical value.

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What is a Classification Problem?

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Predicting a class

In supervised **classification**, each training example has: (x_i, y_i),

where:

- x_i is a feature vector;
- y_i is a class label.

For binary classification: $y_i \in \{0, 1\}$

Examples

Problem	Features	Target class
Email filtering	words, sender, links	spam / not spam
Medical screening	age, symptoms, tests	disease / no disease
Fraud detection	amount, merchant, time	fraud / legitimate
Student success	attendance, grades, activity	pass / fail

> The class label is discrete, even if the model internally computes probabilities.

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Binary Classification

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Positive and negative classes

Binary classification has two possible outcomes.

We usually choose one class as the **positive class**: $y = 1$

The other class is the **negative class**: $y = 0$

Example — Fraud detection

Class	Label
Fraud	1
Legitimate	0

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Important

The choice of positive class affects interpretation of precision, recall, false positives, and false negatives.

> If fraud is the positive class, a false negative means a fraudulent transaction was missed.

Classification Output

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Class label or probability?

A classifier may produce:

- **Class label:** A discrete prediction, such as 0 or 1.
- **Probability:** An estimate of how likely class 1 is.
- **Score:** A numerical value used to make a decision.

Logistic regression

Logistic regression estimates the probability:

$$p = P(y = 1 \mid x)$$

A **decision threshold** converts that probability into a class label.

For the common threshold $t = 0.5$:

$$\hat{y} = \begin{cases} 1, & p \geq 0.5 \\ 0, & p < 0.5 \end{cases}$$

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💡 **Key idea.** Logistic regression predicts a **probability**; the threshold determines the **final class decision**.

Why Not Linear Regression?

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Linear outputs are not bounded probabilities

Ordinary **linear regression** predicts:

$$\hat{y} = w^\top x + b$$

For binary classification, this creates problems:

- Predictions may fall **below 0 or above 1**.
- Outputs cannot always be interpreted as **probabilities**.
- Squared error is possible, but **binary cross-entropy** is the standard logistic-regression loss.

The logistic-regression solution

Apply the **sigmoid function** to the linear score:

$$p = \sigma(w^\top x + b) = \frac{1}{1 + e^{-(w^\top x + b)}}$$

This ensures that $0 < p < 1$.

> 💡 **Key idea.** Logistic regression combines a **linear score** with the **sigmoid function** to produce probabilities suitable for binary classification.

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Part II

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The Logistic Model

Logistic regression transforms a **linear score** into a probability using the **sigmoid function**.

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Linear Score

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Start with a weighted sum

Like linear regression, logistic regression first computes a **linear score**:

$$\boxed{z = w^\top x + b = w_1 x_1 + \cdots + w_d x_d + b}$$

where w_j are feature weights, x_j are input features, and b is the bias.

The score can take **any real value**:

$$z \in (-\infty, \infty)$$

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How should we interpret z?

- $z > 0$: Favors class 1.
- $z < 0$: Favors class 0.
- $z = 0$: Corresponds to probability 0.5 after applying sigmoid.

💡 **Key idea.** The score z is **not yet a probability**. Logistic regression transforms it using the sigmoid function.

Sigmoid Function

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From score to probability

The **sigmoid function** maps a linear score to a probability:

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}, \qquad z = w^\top x + b$$

What happens as the score changes?

Linear score z	Predicted probability p
Large negative	Approaches 0
0	Exactly 0.5
Large positive	Approaches 1

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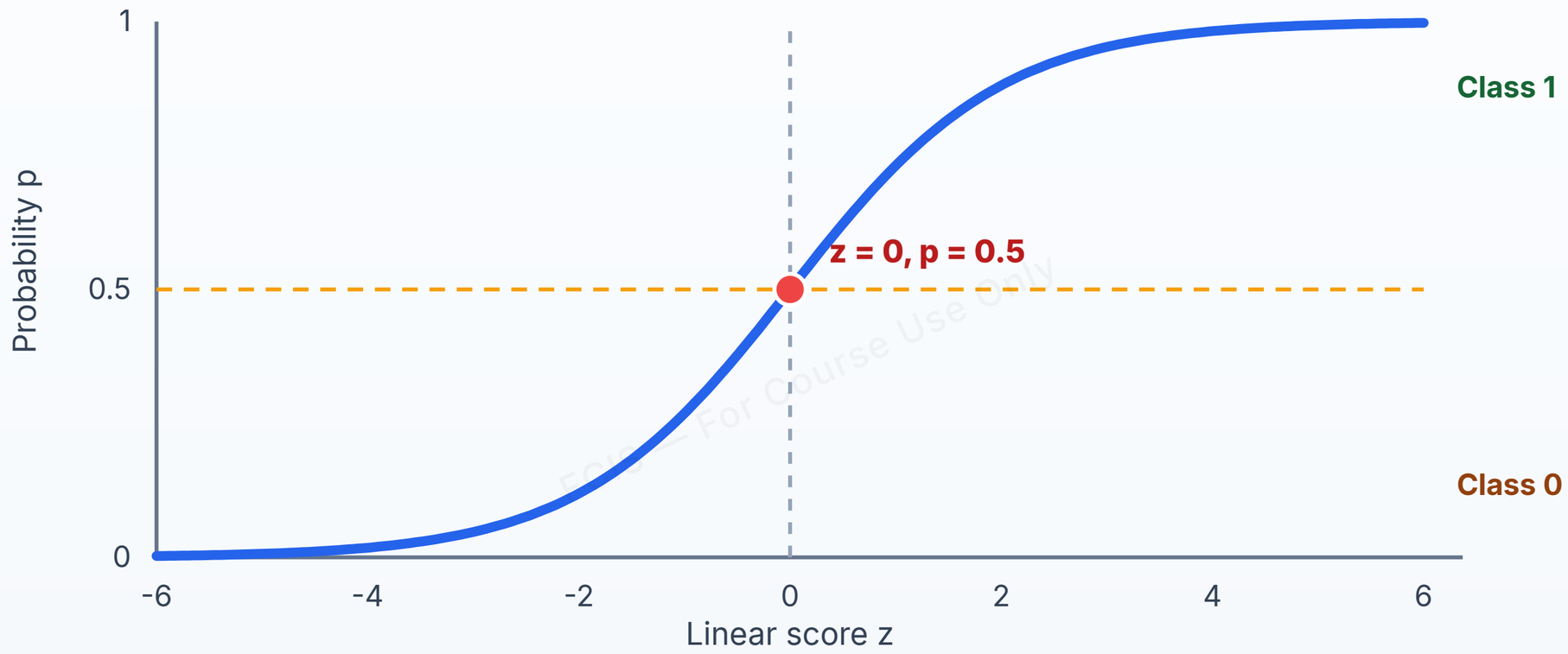
The function is **smooth and increasing**: larger scores always produce larger predicted probabilities.

> **Sigmoid:** A nonlinear function that transforms any real number into the interval $(0, 1)$.

Visualizing the Sigmoid

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How a score becomes a probability



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💡 **Key observation.** The sigmoid is steepest near $z = 0$ and flattens near probabilities **0 and 1**. At the usual threshold of 0.5, the decision changes at $z = 0$.

Sigmoid Interpretation

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What do different scores mean?

The **sign and magnitude** of the linear score determine the predicted probability.

Score z	Probability $\sigma(z)$	Interpretation
-4	0.018	Strongly favors class 0
-1	0.269	Favors class 0
0	0.500	At the 0.5 decision boundary
1	0.731	Favors class 1
4	0.982	Strongly favors class 1

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Why does the curve flatten?

Increasing z from **0 to 2** changes the probability much more than increasing it from **8 to 10**.

Important. A probability near 0 or 1 means the **model is highly confident**, not necessarily correct. Probability calibration should be checked on held-out data.

Worked Forward Pass

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Compute one logistic-regression prediction

Suppose a classifier uses:

$$z = 0.8x_1 - 0.4x_2 + 0.2$$

For a new observation, $x_1 = 2$ and $x_2 = 1$.

Step 1: Compute the linear score

$$z = 0.8(2) - 0.4(1) + 0.2 = \boxed{1.4}$$

Step 2: Apply the sigmoid

$$p = \sigma(1.4) = \frac{1}{1 + e^{-1.4}} \approx \boxed{0.802}$$

Prediction: The model estimates an **80.2% probability of class 1**. With threshold $t = 0.5$, the predicted class is $\boxed{\hat{y} = 1}$.

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Odds and Log-Odds

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What logistic regression is linear in

Let $p = P(y = 1 \mid x)$ be the predicted probability of the positive class.

From probability to log-odds

$$\text{Odds} = \frac{p}{1 - p}$$

$$\text{Log-odds} = \log\left(\frac{p}{1 - p}\right)$$

Logistic regression assumption

$$\boxed{\log\left(\frac{p}{1 - p}\right) = w^\top x + b}$$

Example: If $p = 0.8$, the odds are $0.8/0.2 = 4$, or **4 to 1**, and the log-odds are $\log(4) \approx 1.39$.

💡 **Key idea.** Logistic regression models the **log-odds as a linear function of the features** (plus an intercept), while the predicted **probability is nonlinear**.

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Decision Rule

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Turning probability into a class

For predicted probability $p = P(y = 1 \mid x)$ and threshold t:

$$\hat{y} = \begin{cases} 1, & p \geq t \\ 0, & p < t \end{cases}$$

A common default is $t = 0.5$, but the appropriate threshold depends on the **costs of classification errors**.

Example: Fraud Detection

Suppose a transaction has predicted fraud probability $p = 0.40$.

- $t = 0.50$: Classify as legitimate ($\hat{y} = 0$).
- $t = 0.30$: Flag as potentially fraudulent ($\hat{y} = 1$).

Lowering the threshold flags **more transactions**, typically increasing recall but also increasing false positives.

> 💡 **Key idea.** The threshold controls the trade-off between **missed fraud** and **unnecessary alerts**. Choose it based on validation results and real-world costs.

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Part III

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Decision Boundaries

Logistic regression converts a linear score into a probability. A **decision threshold** determines where the model predicts each class.

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Boundary at Threshold 0.5

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Where predictions switch classes

Logistic regression predicts $p = \sigma(z)$, where $z = w^\top x + b$.

Since $\sigma(0) = 0.5$, at the usual threshold $t = 0.5$:

$$\boxed{w^\top x + b = 0}$$

In two dimensions

The decision boundary is a **straight line**:

$$w_1 x_1 + w_2 x_2 + b = 0$$

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The classification rule is:

- **Class 1:** $w^\top x + b \geq 0$
- **Class 0:** $w^\top x + b < 0$

💡 **Key idea.** With the original features as inputs, logistic regression produces a **linear decision boundary**: a line in 2D or a hyperplane in higher dimensions.

Changing the Threshold

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Same model, different decisions

For logistic regression, $p = \sigma(z)$ with $z = w^\top x + b$.

Because the sigmoid is increasing, for $0 < t < 1$:

$$p \geq t \iff w^\top x + b \geq \log\left(\frac{t}{1-t}\right)$$

Compare two thresholds

Threshold	Decision boundary
$t = 0.50$	$w^\top x + b = 0$
$t = 0.75$	$w^\top x + b = \log(3) \approx 1.099$

Increasing the threshold makes a **positive prediction harder to obtain**.

Important. Changing the threshold shifts the boundary but **does not change the trained weights or bias**. Choose thresholds using validation data and application-specific error costs.

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Worked Boundary Example

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Interpret a two-feature classifier

Suppose the model uses the score:

$$z = 2x_1 + x_2 - 6$$

At threshold $t = 0.5$, the decision boundary is:

$$2x_1 + x_2 - 6 = 0 \quad \Longrightarrow \quad \boxed{x_2 = 6 - 2x_1}$$

Classify the point $(2, 3)$

$$z = 2(2) + 3 - 6 = 1, \qquad p = \sigma(1) \approx 0.731$$

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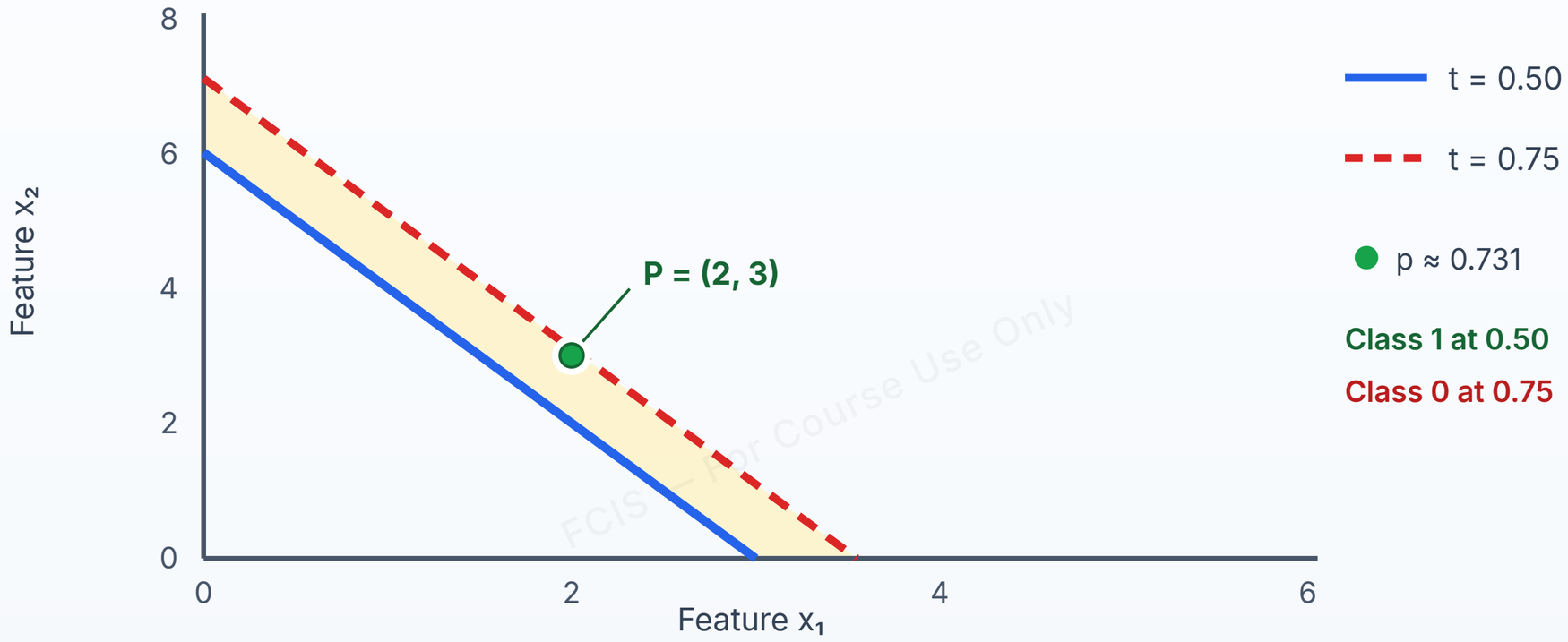
Threshold	Predicted class
$t = 0.50$	**Class 1** $(0.731 \geq 0.50)$
$t = 0.75$	**Class 0** $(0.731 < 0.75)$

💡 **Key idea.** The probability stays at **0.731**. Only the **decision threshold** changes.

Visualizing Threshold-Dependent Boundaries

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Same model, parallel decision boundaries



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💡 **Key observation.** The shaded region contains points classified as **class 1 at** $t = 0.50$ but **class 0 at** $t = 0.75$. The model's weights and bias remain unchanged.

Effect of Coefficients

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Direction and strength

In logistic regression, the linear score is:

$$z = w_1 x_1 + \cdots + w_p x_p + b$$

Holding all other features fixed:

Coefficient	Effect of increasing x_j
$w_j > 0$	**Increases** the score and predicted probability
$w_j < 0$	**Decreases** the score and predicted probability
$w_j = 0$	Has **no direct effect** on the score

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Interpreting magnitude

A one-unit increase in x_j multiplies the **odds** by e^{w_j}, holding the other features fixed.

> 💡 **Key idea.** Coefficient signs indicate **direction**. Magnitudes depend on feature units, so compare their sizes cautiously; standardization can help.

Linear Separability

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When a straight boundary is not enough

With the original features, logistic regression uses a **linear decision boundary**.

It can still classify overlapping groups, but a straight boundary may not capture **curved or more complex patterns**.

How can we model nonlinear boundaries?

- **Polynomial features:** Include terms such as x_1^2 or x_2^2.
- **Interaction features:** Include terms such as $x_1 x_2$.
- **Alternative models:** Consider decision trees or neural networks when suitable.

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Important distinction

A model can be **linear in its supplied features** while having a **nonlinear boundary in the original feature space**.

> 💡 **Key idea.** Feature engineering can make logistic regression more flexible **without changing the logistic regression algorithm**.

Part IV

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Binary Cross-Entropy Loss

For classification, a good loss should **reward accurate probabilities** and **strongly penalize confident mistakes**.

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Why Not Squared Error?

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Choosing a loss for classification

Squared error can measure probability errors, but **binary cross-entropy** is the standard loss for logistic regression.

We want a loss that:

- Rewards assigning **high probability to the correct class**.
- Strongly penalizes **confident incorrect predictions**.
- Matches the **Bernoulli probability model** underlying logistic regression.
- Works naturally with the **sigmoid function** during optimization.

Binary cross-entropy is also called **log loss**.

> **Important.** A low log loss encourages good probability predictions, but it does **not guarantee perfect calibration**. Predicted probabilities should still be evaluated on unseen data.

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Binary Cross-Entropy

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Loss for one observation

Let $y \in \{0, 1\}$ be the true class and $p = P(y = 1 \mid x)$ the predicted probability.

The **binary cross-entropy loss** is:

$$\boxed{L(y, p) = -\left[y \log(p) + (1 - y) \log(1 - p)\right]}$$

Two possible cases

True class	Loss	What reduces the loss?
$y = 1$	$-\log(p)$	Increasing p toward 1
$y = 0$	$-\log(1 - p)$	Decreasing p toward 0

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💡 **Key idea.** One formula handles both classes by measuring how much probability the model assigns to the **true outcome**.

Interpreting Log Loss

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Confident mistakes are expensive

Consider the loss for different predicted probabilities:

True class y	Predicted $p = P(y = 1 \mid x)$	Log loss
1	0.90	0.105
1	0.60	0.511
1	0.10	2.303
0	0.10	0.105
0	0.90	2.303

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For example, if $y = 1$ but $p = 0.10$:

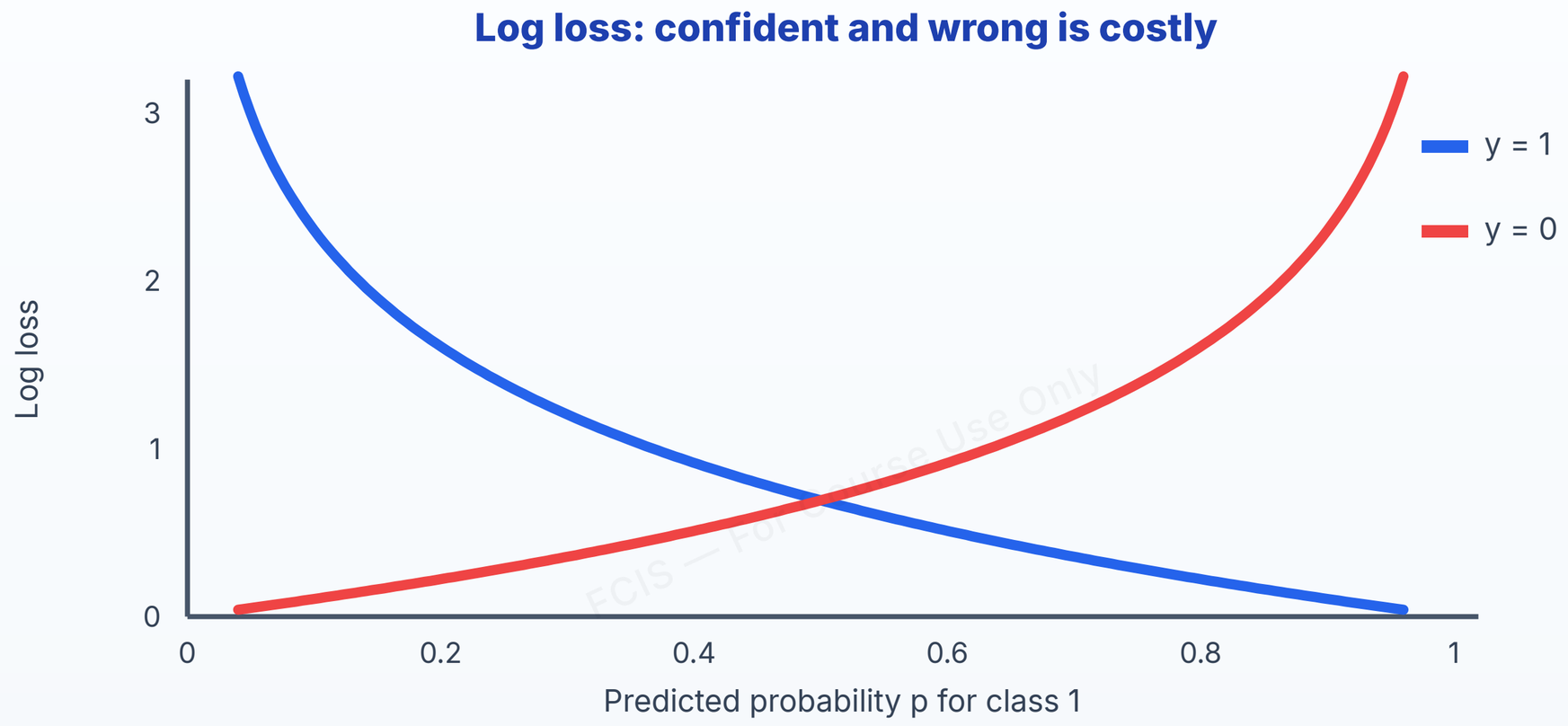
$$L = -\log(0.10) \approx 2.303$$

💡 **Key observation.** A **confident correct prediction** has low loss, while a **confident wrong prediction** has high loss. Log loss grows without bound as the probability assigned to the true class approaches zero.

Visualizing Binary Cross-Entropy

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How loss changes with predicted probability



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💡 For a truly positive case ($y = 1$), loss falls as p rises; for a truly negative case ($y = 0$), it rises as p rises. Both curves **penalize confidently wrong predictions sharply**.

Dataset Loss

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Average loss over training examples

For n training observations, logistic regression minimizes the **average binary cross-entropy**:

$$J(w, b) = -\frac{1}{n} \sum_{i=1}^{n} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where the predicted probability for observation i is:

$$p_i = \sigma(w^\top x_i + b)$$

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Training objective

Find the weights w and bias b that **minimize** $J(w, b)$.

> 💡 **Key idea.** Training logistic regression means adjusting its parameters to assign **higher probabilities to the observed classes**, thereby reducing average log loss.

Part V

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Learning Logistic Regression

Training adjusts the **weights and bias** to make predicted probabilities agree more closely with the observed labels.

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Forward Pass

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From features to probability to loss

For one training example, logistic regression follows three steps:

Step	Computation
1. Linear score	$z = w^\top x + b$
2. Predicted probability	$p = \sigma(z)$
3. Binary cross-entropy	$L = -[y \log(p) + (1 - y) \log(1 - p)]$

What happens next?

Compute **gradients** of the loss and use them to adjust the weights and bias.

> 💡 **Learning cycle:** Forward pass → Loss → Gradients → Parameter update.
>
> This same pattern appears in **neural-network training**.

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Gradient for One Example

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A remarkably simple result

For logistic regression with binary cross-entropy, the derivative of the loss with respect to the linear score is:

$$\boxed{\frac{\partial L}{\partial z} = p - y}$$

Using the **chain rule**, the parameter gradients are:

$$\boxed{\nabla_w L = (p - y)x, \qquad \frac{\partial L}{\partial b} = p - y}$$

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For each individual weight, $\frac{\partial L}{\partial w_j} = (p - y)x_j$.

Interpreting the error signal

- $y = 1$: $p - y < 0$, so the update should increase the score.
- $y = 0$: $p - y > 0$, so the update should decrease the score.

> 💡 **Key idea.** The quantity $p - y$ is the error signal that drives the gradient update.

Gradient Descent Update

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Adjust parameters to reduce loss

Gradient descent moves parameters in the **opposite direction of the gradient**.

For learning rate $\alpha > 0$:

$$\boxed{\begin{aligned} w &\leftarrow w - \alpha(p - y)x \\ b &\leftarrow b - \alpha(p - y) \end{aligned}}$$

For an individual weight:

$$w_j \leftarrow w_j - \alpha(p - y)x_j$$

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What does the update do?

- **Positive example ($y = 1$):** Increases its predicted probability.
- **Negative example ($y = 0$):** Decreases its predicted probability.
- **Learning rate α:** Controls the size of the update.

💡 **Key idea.** Each update adjusts the model to assign a **higher probability to the observed class** for the current example.

Worked Training Step

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Step 1: Forward pass

Consider one training example with label $y = 1$.

The features and initial parameters are:

$$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad w = \begin{bmatrix} 0.5 \\ -0.25 \end{bmatrix}, \quad b = 0.1$$

Compute the score, probability, and loss

$$z = 0.5(2) - 0.25(1) + 0.1 = \boxed{0.85}$$

$$p = \sigma(0.85) \approx \boxed{0.701}$$

$$L = -\log(p) \approx \boxed{0.356}$$

Interpretation: The true class is **1**, but the model assigns it only about **70.1% probability**. Training should increase that probability.

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Worked Training Step

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Step 2: Backward pass and update

From the forward pass, $p \approx 0.701$ and $y = 1$.

Calculate gradients

$$p - y = 0.701 - 1 = -0.299$$

$$\nabla_w L = (p - y)x = \begin{bmatrix} -0.598 \\ -0.299 \end{bmatrix}$$

$$\frac{\partial L}{\partial b} = -0.299$$

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Update with learning rate $\alpha = 0.1$

$$w_1 \leftarrow 0.5 - 0.1(-0.598) = \boxed{0.5598}$$

$$w_2 \leftarrow -0.25 - 0.1(-0.299) = \boxed{-0.2201}$$

$$b \leftarrow 0.1 - 0.1(-0.299) = \boxed{0.1299}$$

💡 **Key observation.** Both weight updates and the bias update are determined by the **error signal** $p - y$ and the learning rate.

After the Update

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Did the prediction improve?

Using the updated parameters and the **same observation**:

$$\begin{aligned} z_{\text{new}} &= 0.5598(2) - 0.2201(1) + 0.1299 \\ &= \boxed{1.0294} \end{aligned}$$

The new probability is $p_{\text{new}} = \sigma(1.0294) \approx 0.737$.

Before versus after

Quantity	Before	After
Linear score z	0.850	1.029
Probability p	0.701	0.737
Log loss L	0.356	0.305

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Result: The probability assigned to the **correct class increased**, and the example's loss **decreased**.

This improvement is for **one training example**. An individual stochastic update does not guarantee lower loss across the entire dataset.

Batch Gradients

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Average gradients across observations

For n training examples, let: $p_i = \sigma(w^\top x_i + b)$

The gradients of the **average binary cross-entropy** are:

$$\boxed{\begin{aligned} \nabla_w J &= \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)x_i \\ \frac{\partial J}{\partial b} &= \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i) \end{aligned}}$$

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Gradient descent then uses:

$$w \leftarrow w - \alpha \nabla_w J, \qquad b \leftarrow b - \alpha \frac{\partial J}{\partial b}$$

Three ways to compute updates

- **Batch:** Use all training examples.
- **Mini-batch:** Use a subset of examples.
- **Stochastic:** Use one example at a time.

Logistic Regression Training Algorithm

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From repeated updates to a trained model

Mini-batch gradient descent — pseudocode

```
initialize weights w and bias b

for each epoch:
    for each mini-batch (X, y):
        p = sigmoid(X @ w + b)
        error = p - y

        grad_w = X.T @ error / len(y)
        grad_b = mean(error)

        w = w - alpha * grad_w
        b = b - alpha * grad_b
```

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Training and validation decisions

During training	Using validation data
Update **weights and bias**	Select learning rate, epochs, and regularization
Minimize the training loss	Choose a decision threshold and compare models

Important. The decision threshold is **not learned by gradient descent**. Select it using validation data, keep the final test set untouched, and fit preprocessing statistics on **training data only**.

Part VI

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Thresholds, Decisions, and Probability Quality

A classifier produces **probabilities**, but using those predictions responsibly requires appropriate **decision thresholds** and reliable probability estimates.

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Thresholds

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The default is not always best

For predicted probability $p = P(y = 1 \mid x)$, classify an observation as positive when:

$$\boxed{p \geq t \implies \hat{y} = 1}$$

A common default is $t = 0.5$, but changing the threshold changes which observations are flagged.

Threshold change	Effect on decisions
Lower t	More positive predictions; recall cannot decrease, but false positives may increase
Higher t	Fewer positive predictions; false positives cannot increase, but false negatives may increase

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Precision is not guaranteed to change monotonically as the threshold changes.

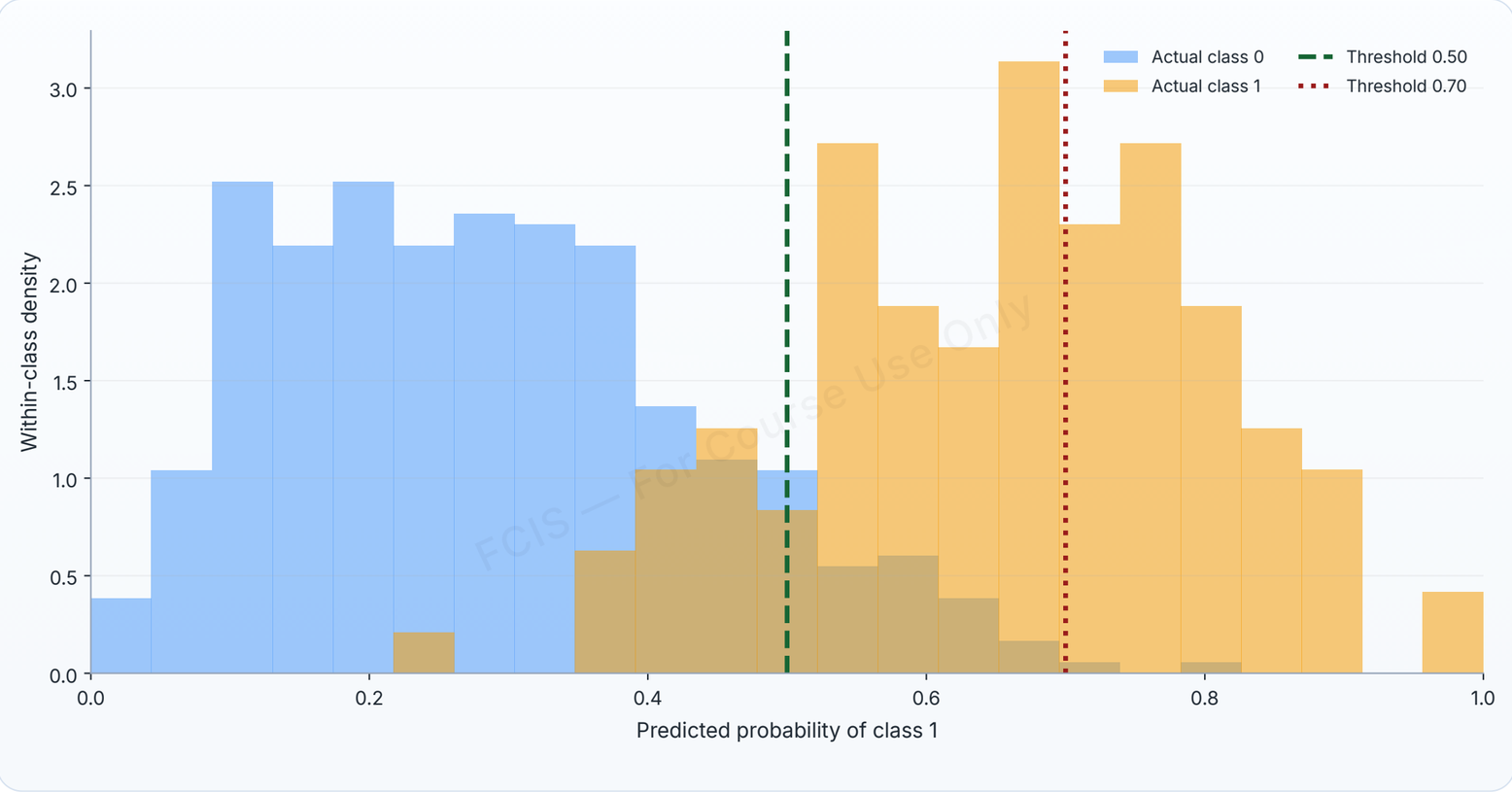
> **Disease screening:** A lower threshold may help identify more potential cases, accepting additional false alarms to reduce missed cases.

Overlapping Class Scores

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Why thresholds create different error trade-offs

These **synthetic scores** overlap across the two actual classes. Both thresholds are applied to the **same predictions**.



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💡 **Key observation.** Moving the threshold from **0.50 to 0.70** flags fewer observations, potentially reducing false alarms but missing more actual positives. The underlying scores do not change.

Threshold Example

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Fraud detection

Suppose a fraud-detection model predicts the following probabilities:

Transaction	$P(\text{fraud})$	Flag at $t = 0.50$?	Flag at $t = 0.20$?
A	0.91	Yes	Yes
B	0.64	Yes	Yes
C	0.48	No	Yes
D	0.22	No	Yes
E	0.07	No	No

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What changes?

- $t = 0.50$: Flag transactions **A and B**.
- $t = 0.20$: Flag transactions **A, B, C, and D**.

Important. Lowering the threshold creates more alerts. Without **true transaction labels**, we cannot determine how many additional alerts are genuine fraud or false positives.

Confusion Matrix Preview

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Understanding classification errors

Once true labels are available, we can compare **predicted decisions** with **actual outcomes**.

	Predicted positive	Predicted negative
Actual positive	True positive (TP)	False negative (FN)
Actual negative	False positive (FP)	True negative (TN)

Example: Fraud detection

- **False negative:** A fraudulent transaction is missed.
- **False positive:** A legitimate transaction is incorrectly flagged.

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These errors may have **different financial and operational costs**.

💡 **Coming in Lecture 5:** We will use the confusion matrix to calculate **accuracy, precision, recall, specificity, and** F_1, and evaluate decisions based on error costs.

Probability Calibration

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Does 0.8 really mean 80%?

Suppose a classifier assigns a predicted probability near **0.80** to many observations.

If the predictions are **well calibrated**, approximately **80% of those observations should actually belong to class 1**.

> **Probability calibration** means that, among observations receiving similar predicted probabilities, the observed positive fraction approximately matches those probabilities.

Why does calibration matter?

Reliable probabilities support:

- **Risk estimation:** Medical, credit, and fraud-related decisions.
- **Threshold selection:** Comparing the consequences of different policies.
- **Resource allocation:** Prioritizing cases according to estimated risk.

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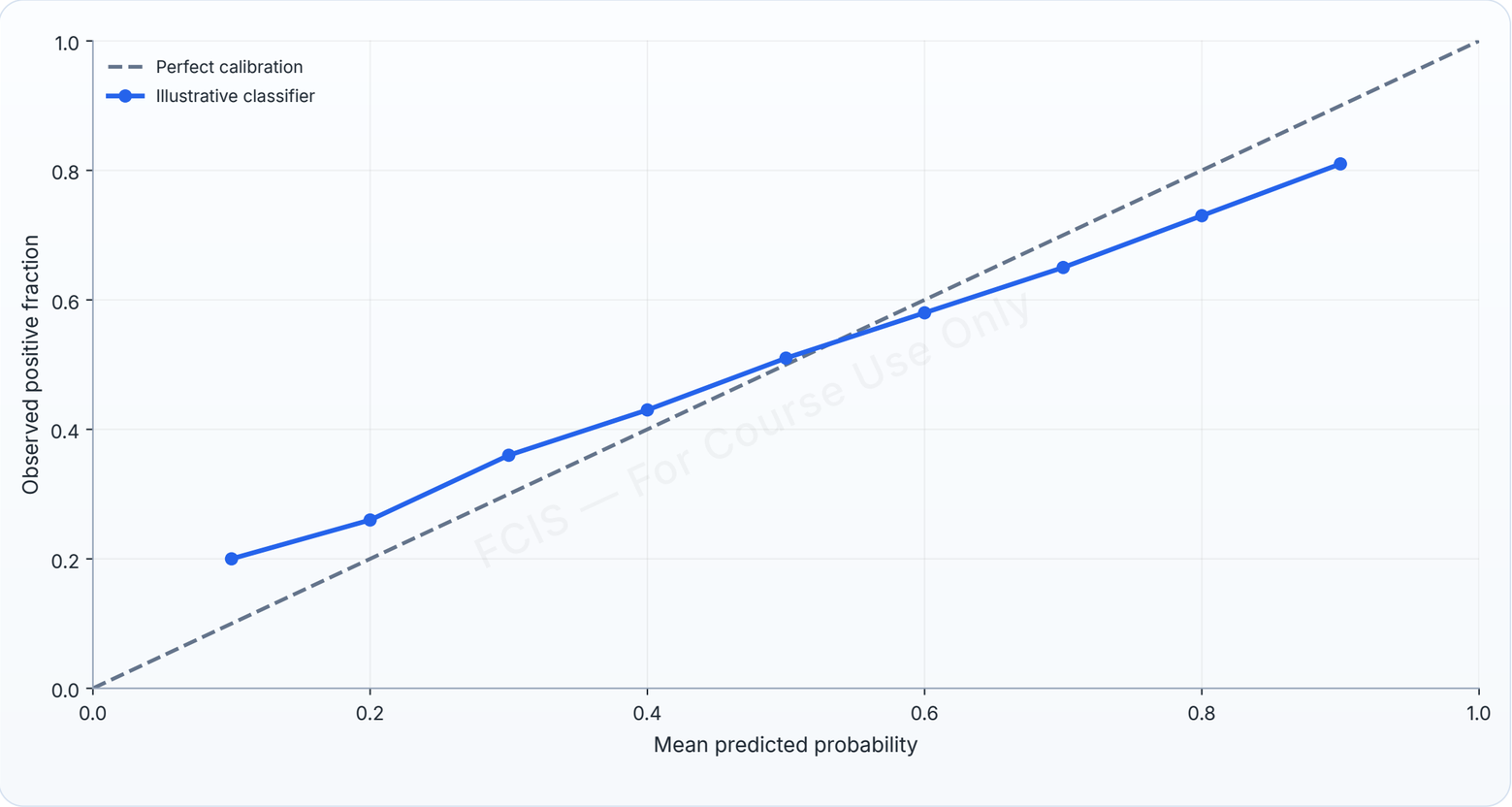
> 💡 **Key distinction.** Classification accuracy evaluates **thresholded decisions**; calibration evaluates whether the **predicted probabilities are trustworthy**.

Probability Calibration

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Comparing predictions with observed outcomes

Each point represents a **hypothetical probability bin**. A well-calibrated model should lie near the dashed diagonal.



💡 **Reading the plot:** At a predicted probability of **0.80**, the observed positive fraction is approximately **0.73**. The classifier is **overconfident in this illustrative bin**.

Class Imbalance

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Why accuracy can be misleading

Suppose only **1% of transactions are fraudulent**.

Consider a classifier that predicts **every transaction as legitimate**.

Evaluation measure	Result
Accuracy	**99%**
Fraud recall	**0%**

The classifier appears highly accurate, yet **detects none of the fraud**.

What should we evaluate instead?

- **Recall:** How many actual fraud cases are detected?
- **Precision:** How many flagged cases are genuine fraud?
- **Error costs:** What is the impact of missed fraud versus false alarms?
- **Threshold choice:** Which decision rule best supports the application?

Key lesson. High accuracy does **not** guarantee a useful classifier, especially when one class is rare.

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Part VII

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Interpreting Logistic Regression

Logistic regression is often attractive because we can interpret how features influence the predicted **log-odds, probabilities, and decisions**.

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Coefficient Interpretation

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Direction of association

Logistic regression models the **log-odds** of the positive class:

$$\log\left(\frac{p}{1-p}\right) = w^\top x + b$$

Holding all other features fixed:

- $w_j > 0$: Increasing x_j increases the log-odds and predicted probability of class 1.
- $w_j < 0$: Increasing x_j decreases the log-odds and predicted probability of class 1.
- $w_j = 0$: No direct effect on the model's prediction.

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Example: Predicting loan default

Suppose $y = 1$ **means default** and a feature is:

Debt-to-income ratio (DTI) = Monthly debt payments / Gross monthly income.

A **positive DTI coefficient** means higher DTI is associated with a higher predicted probability of default, holding other features fixed.

> 💡 **Key idea.** Coefficient signs indicate the **direction of association**, not causation. Their magnitudes depend on feature units and scaling.

Odds Ratio

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Exponentiating coefficients

For a one-unit increase in feature x_j, holding all other features fixed, the **odds of class 1** are multiplied by:

$$\boxed{\text{Odds ratio} = e^{w_j}}$$

Worked example

Suppose $w_j = 0.7$.

$$e^{0.7} \approx \boxed{2.01}$$

This means a one-unit increase in x_j **approximately doubles the odds** of class 1.

Odds ratio	Interpretation
Greater than 1	Odds increase
Equal to 1	Odds remain unchanged
Less than 1	Odds decrease

Important. An odds ratio of **2** does **not** mean the probability doubles. Interpretation also depends on what a **one-unit change** in the feature represents.

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Interpreting Encoded Categories

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Coefficients depend on the reference category

Suppose **Learning Mode** has three categories, with **individual learning** chosen as the reference.

Category	`mode_group`	`mode_online`
Individual (reference)	0	0
Group	1	0
Online	0	1

The model includes coefficients for the two encoded features.

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How do we interpret them?

- `mode_group`: Compares group learning with **individual learning**.
- `mode_online`: Compares online learning with **individual learning**.

For example, a **positive** `mode_group` **coefficient** indicates higher predicted log-odds of class 1 for group learning than for individual learning, holding other features fixed.

> 💡 **Key idea.** Categorical coefficients are interpreted **relative to the omitted reference category**, not independently of it.

Regularized Logistic Regression

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Controlling coefficient magnitude

As in Lecture 3, **regularization** controls model complexity. For logistic regression, we add a penalty to the **binary cross-entropy objective**.

L2-regularized objective

$$J_{\text{reg}}(w, b) = J(w, b) + \lambda \sum_{j=1}^{d} w_j^2$$

Here, $\lambda \geq 0$ controls the **regularization strength**; the bias is not penalized in this formulation.

Why use regularization?

- **Reduce overfitting** by discouraging overly large coefficients.
- **Stabilize estimates**, especially with correlated features.
- **Handle high-dimensional data**, such as text features.
- **Improve generalization** when the strength is chosen appropriately.

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💡 **Key idea.** L2 regularization shrinks coefficients toward zero. Stronger regularization reduces model flexibility, so choose its strength using **validation data**.

Part VIII

FCIS — Lecture Use Only

Preparing Features for Logistic Regression

Logistic regression is mathematically simple, but its performance and interpretation depend on how the **input features are represented**.

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Numerical Features

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Why scaling can help

Feature scaling is often beneficial for logistic regression because it can:

- Improve the convergence of **gradient-based optimization**.
- Make **regularization** act more consistently across features.
- Make coefficients more comparable when features are standardized.
- Improve numerical conditioning when features have very different scales.

Common scaling methods

Method	Transformation
Standardization	$x_{\text{std}} = \dfrac{x - \mu}{\sigma}$
Min-max scaling	$x' = \dfrac{x - x_{\min}}{x_{\max} - x_{\min}}$

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Standardization centers a feature around zero; min-max scaling maps training values to a chosen range, typically $[0, 1]$.

Avoid data leakage. Estimate scaling parameters from the **training data only**. Apply the same transformation to validation and test data.

Categorical Features

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Convert categories into numerical inputs

Logistic regression requires **numerical features**. For nominal categories, a common approach is **one-hot encoding**.

Example: City

With **Madinah as the reference category**:

City	`city_riyadh`	`city_jeddah`
Madinah (reference)	0	0
Riyadh	1	0
Jeddah	0	1

The coefficients compare **Riyadh** and **Jeddah** with **Madinah**, holding other features fixed.

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Why omit one category?

With an intercept, using one reference category avoids redundant indicators and gives the coefficients a clear interpretation.

> **Important.** High-cardinality features can create many columns. Fit the encoder using **training data**, and define how to handle previously unseen categories.

Nonlinear Patterns

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Feature engineering increases flexibility

Logistic regression produces a **linear decision boundary in the features supplied to the model**.

When the original relationships are nonlinear, we can create more informative features.

Common transformations

- **Polynomial features:** Include x_1^2 or x_2^2.
- **Interaction features:** Capture combined effects of two variables.
- **Log transformations:** Represent certain skewed numerical features.
- **Domain-specific features:** Use meaningful ratios, rates, or combinations.

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Example: Interaction

$$x_{\text{interaction}} = x_1 x_2$$

The model can now assign a coefficient to the **combined effect** of x_1 and x_2.

> 💡 **Key idea.** Logistic regression remains **linear in its engineered features**, but its decision boundary can become **nonlinear in the original feature space**.

Part IX

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Beyond Binary Classification

Many classification tasks involve **more than two classes**. Logistic regression can be extended to predict among multiple possible outcomes.

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From Binary to Multiclass

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Predicting more than two classes

In **multiclass classification**, each observation belongs to **one of K mutually exclusive classes**.

Examples

Application	Possible classes
Digit recognition	0, 1, 2, ..., 9
Email routing	Academic, finance, IT, administration
Primary diagnosis	Disease A, disease B, disease C
Sentiment analysis	Negative, neutral, positive

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Two common approaches

1. **One-vs-rest (OvR):** Train a separate binary classifier for each class.
2. **Softmax regression:** Learn class scores jointly and convert them into a probability distribution.

> 💡 **Key idea.** Both approaches extend classification beyond two labels, but they differ in **how class scores and probabilities are produced**.

One-vs-Rest

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One binary classifier for each class

For K classes, **one-vs-rest (OvR)** trains K binary classifiers.

Each classifier answers:

> **Does this observation belong to class k, or to any other class?**

Example: Sentiment classification

Train three classifiers:

- **Positive** versus not positive.
- **Neutral** versus not neutral.
- **Negative** versus not negative.

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At prediction time, select the class with the **highest classifier score**.

Strengths and limitation

- **Strength:** Simple extension of binary logistic regression.
- **Limitation:** The classifiers are trained independently, so their probability estimates do **not necessarily sum to 1**.

> 💡 **Key idea.** OvR solves a multiclass problem using **multiple binary classification problems**.

Softmax Regression

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A probability distribution over classes

For each class k, compute a **linear score**:

$$z_k = w_k^\top x + b_k$$

The **softmax function** converts these scores into class probabilities:

$$\boxed{P(y = k \mid x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}}$$

The probabilities are **nonnegative and sum to 1**.

Example: Three classes

If the class scores are $(2, 1, 0)$, softmax gives approximately:

$$(0.665,\ 0.245,\ 0.090)$$

The model predicts **class 1**, which has the highest probability.

💡 **Key idea.** Softmax regression learns a **joint multiclass model**, typically trained with **multiclass cross-entropy loss**. Softmax is also widely used in neural-network output layers.

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Part X

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Applied Mini-Case: Student Success

How can **logistic regression** turn student data into **risk estimates** that support timely, responsible academic interventions?

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Mini-Case: Student Success Prediction

For Course Use Only

Define the prediction problem

A university wants to identify students who may need **early academic support**.

Features available during the course

Feature	Type	Preparation
Attendance percentage	Numerical	Scaling
Missed labs	Count	Check range and scale if useful
Previous GPA	Numerical	Scaling
Learning mode	Categorical	One-hot encoding
LMS activity score	Numerical	Scaling

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Target: Final course outcome

- $y = 1$: Student failed the course.
- $y = 0$: Student passed the course.

The model estimates $P(y = 1 \mid x)$, interpreted as the predicted risk of course failure.

Responsible use. Use only features available at prediction time. Risk estimates should guide **supportive outreach**, not automatic punitive decisions.

Mini-Case: Model Output

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From predicted risk to action

Suppose the model produces these **illustrative failure probabilities**:

Student	$P(\text{failure})$	$t = 0.50$	$t = 0.25$
A	0.72	Flag	Flag
B	0.46	—	Flag
C	0.28	—	Flag
D	0.12	—	—

A **flag** means the student may be considered for additional support.

Questions

1. Which threshold flags more students?
2. Which threshold could reduce **missed at-risk students**?
3. What is the possible cost of flagging more students?

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From predicted risk to action

Suppose the model produces these **illustrative failure probabilities**:

Student	$P(\text{failure})$	$t = 0.50$	$t = 0.25$
A	0.72	Flag	Flag
B	0.46	—	Flag
C	0.28	—	Flag
D	0.12	—	—

A **flag** means the student may be considered for additional support.

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Questions

1. Which threshold flags more students?
2. Which threshold could reduce **missed at-risk students**?
3. What is the possible cost of flagging more students?

Discussion: Lowering the threshold from **0.50 to 0.25** adds students **B and C**. This can reduce missed cases, but may also increase unnecessary alerts. We need actual outcomes to evaluate the trade-off.

Mini-Case: Check the Decisions

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Illustrative validation outcomes

Suppose we know the **actual course outcomes** for these same four students.

Student	Predicted risk	True label y	$t = 0.50$	$t = 0.25$
A	0.72	1	Positive	Positive
B	0.46	0	Negative	Positive
C	0.28	1	Negative	Positive
D	0.12	0	Negative	Negative

Compare the thresholds

Threshold	TP	FN	FP	TN	Precision	Recall
0.50	1	1	0	2	100%	50%
0.25	2	0	1	1	67%	100%

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💡 **Interpretation.** The lower threshold identifies student **C**, who would otherwise be missed, but also flags student **B**, who ultimately passes.

These **four fictional cases** demonstrate the calculation, not real model performance. Select thresholds using representative validation data and the costs and benefits of interventions.

Mini-Case: Reasoning Questions

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Evaluate the complete pipeline

1. Why is predicting **course failure** a classification task?
2. Why does logistic regression produce a **probability** before a decision?
3. When might a threshold **below 0.5** be appropriate?
4. How could features measured **after the prediction date** cause leakage?
5. Why should we evaluate the model on **held-out students or later cohorts**?
6. How do the consequences of **false positives and false negatives** differ?

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💡 **Key lesson.** A useful student-success model needs more than accurate predictions: it requires **reliable validation, appropriate thresholds, and responsible decisions**.

Practice & Consolidation

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Logistic Regression: Activities and Common Mistakes

Apply the complete pipeline: **linear score → probability → loss → decision → evaluation**.

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Activity: Compute a Probability

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One forward pass

A logistic-regression model uses:

$$z = 1.2x_1 - 0.8x_2 - 0.4$$

For a new observation, $x_1 = 1$ and $x_2 = 2$.

Tasks

1. Calculate the **linear score** z.
2. Compute the probability $p = \sigma(z)$.
3. Predict the class using **threshold** $t = 0.5$.

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One forward pass

A logistic-regression model uses:

$$z = 1.2x_1 - 0.8x_2 - 0.4$$

For a new observation, $x_1 = 1$ and $x_2 = 2$.

Tasks

1. Calculate the **linear score** z.
2. Compute the probability $p = \sigma(z)$.
3. Predict the class using **threshold** $t = 0.5$.

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Solution

$$z = 1.2(1) - 0.8(2) - 0.4 = \boxed{-0.8}$$
$$p = \sigma(-0.8) \approx \boxed{0.310}$$

Since $p < 0.5$, the predicted class is $\boxed{\hat{y} = 0}$.

Activity: Compute Log Loss

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Compare correct and incorrect predictions

Compute the **binary cross-entropy loss** for each observation.

Case	True label y	Predicted p	Loss
A	1	0.90	?
B	1	0.20	?
C	0	0.10	?
D	0	0.80	?

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Questions

1. Which cases have the **lowest loss**?
2. Which predictions are **confidently wrong**?
3. Why does log loss penalize those errors strongly?

Compare correct and incorrect predictions

Compute the **binary cross-entropy loss** for each observation.

Case	True label y	Predicted p	Loss
A	1	0.90	?
B	1	0.20	?
C	0	0.10	?
D	0	0.80	?

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Questions

1. Which cases have the **lowest loss**?
2. Which predictions are **confidently wrong**?
3. Why does log loss penalize those errors strongly?

Solution: $L_A = L_C = -\log(0.9) \approx \mathbf{0.105}$; $L_B = L_D = -\log(0.2) \approx \mathbf{1.609}$.

A and C are correct with high confidence. **B and D** assign low probability to the true class, so they incur much higher loss.

Activity: Threshold Choice

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Medical Screening

A screening model predicts the probability of disease:

Patient	P1	P2	P3	P4	P5
Probability	0.83	0.58	0.41	0.27	0.09

Questions

1. Who is flagged at $t = 0.50$?
2. Who is flagged at $t = 0.25$?
3. Which threshold produces **fewer alerts**?
4. Which might be preferred if **missing disease is costly**?

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Medical Screening

A screening model predicts the probability of disease:

Patient	P1	P2	P3	P4	P5
Probability	0.83	0.58	0.41	0.27	0.09

Questions

1. Who is flagged at $t = 0.50$?
2. Who is flagged at $t = 0.25$?
3. Which threshold produces **fewer alerts**?
4. Which might be preferred if **missing disease is costly**?

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Solution: At $t = 0.50$, flag **P1, P2**; at $t = 0.25$, flag **P1–P4**.

The **higher threshold** produces fewer alerts. The **lower threshold** may reduce missed cases but increase false alarms. Choose using validation results and clinical costs.

Common Mistakes

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Logistic regression pitfalls

Mistake	Why it matters
Treating probability as certainty	Predictions contain uncertainty
Using accuracy alone for imbalanced data	Important positive cases may be missed
Ignoring scaling when regularizing	Feature scales affect penalties and optimization
Misinterpreting encoded categories	Coefficients depend on the reference category
Tuning thresholds on the test set	Produces biased final performance estimates
Always using threshold 0.5	Ignores application-specific error costs
Using future information as a feature	Creates data leakage
Confusing the name with the task	Logistic regression is commonly used for classification

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💡 **Key lesson.** Reliable logistic regression requires **appropriate features, sound evaluation, and careful interpretation of probabilities and decisions**.

Mini-Quiz

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Check your understanding

1. What probability does **logistic regression** estimate?
2. Why do we apply the **sigmoid** function?
3. What defines the decision boundary at $t = 0.5$?
4. Why is **log loss** preferred over MSE for logistic regression?
5. What is the gradient error signal for one example?
6. Why can lowering a threshold improve **recall**?
7. Why can **class imbalance** make accuracy misleading?
8. How do **one-vs-rest** and **softmax** differ?

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Check your understanding

1. What probability does **logistic regression** estimate?
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8. How do **one-vs-rest** and **softmax** differ?

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Suggested answers: $P(y = 1 \mid x)$; sigmoid maps scores to probabilities; $w^\top x + b = 0$; log loss penalizes confident wrong probabilities; $p - y$; more positives are flagged; a majority-class classifier can get high accuracy; one-vs-rest trains binary classifiers while softmax models a multiclass probability distribution.

Lecture 4 Summary

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Key ideas to retain

Model and Learning

- **Logistic regression** is primarily a classification model.
- It computes a **linear score**: $z = w^\top x + b$.
- **Sigmoid** converts the score into $p = \sigma(z)$.
- **Binary cross-entropy** measures probability prediction errors.
- The gradient error signal is $p - y$.
- **Gradient descent** updates weights and bias to reduce loss.

Decisions and Interpretation

- A **threshold** converts predicted probabilities into class labels.
- The default **0.5 threshold** is not always optimal.
- Coefficients describe changes in **log-odds**.
- **Scaling and encoding** influence training and interpretation.
- **Imbalance and error costs** affect how models should be evaluated.
- **One-vs-rest and softmax** extend classification to multiple classes.

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End-of-Lecture Check

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Five questions to test your understanding

1. How does logistic regression transform a **linear score** into a probability?
2. Why does **binary cross-entropy** penalize confident wrong predictions strongly?
3. What happens to the **decision boundary** when the threshold changes without retraining?
4. Why might a **medical screening model** use a threshold below 0.5?
5. Why do **feature scaling and categorical encoding** matter when interpreting coefficients?

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💡 **Preparation for Lecture 5.** Review **TP, TN, FP, FN, precision, recall,** F_1, validation and test sets, and the trade-off between **different types of prediction errors**. Next, we study how to **evaluate models and select decisions systematically**.

Suggested Reading

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Primary and supporting references

Primary reference

G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.

- Review sections on classification, logistic regression, odds, probabilities, and classifier interpretation.

Supporting references

A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly, 2022.

- Review classification, logistic regression, decision thresholds, and practical Scikit-Learn workflows.

K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.

- Use selected sections for a deeper probabilistic view of logistic models and classification losses.

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> 💡 When reading, focus on the connection between probability prediction, loss, gradient update, and decision threshold.